FedHealth: Collaborative Deep Learning Platform for ICU Mortality Prediction in Medical Institutions

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Abstract

Collaborative deep learning modules that utilize Federated Learning are needed to combat data scarcity, quality, and privacy. Federated Deep Learning allows collaborative model training by multiple institutions, without exchanging raw data, thereby conserving data privacy while using diverse datasets. This work proposes an Intensive Care Unit mortality prediction module profiles individuals with a high chance of mortality in the Intensive Care Unit based on their medical history of Prostate Cancer and related comorbidities and medication. It aims to provide timely insights into medical care, ultimately improving patient survival rates. It also presents a comparative analysis between the performance of Federated Learning algorithms like FedAvg and FedProx for the aforementioned objectives, and addresses the challenge of inaccessible data and privacy concerns prevalent in healthcare settings.

Keywords: Federated Learning, Deep Learning, ICU Mortality Prediction, FedAvg, FedProx, Hyperparameter Tuning

1. Introduction

The integration of advanced healthcare technologies has significantly enhanced patient healthcare and streamlined medical processes. However, many advanced algorithms face constraints due to the lack of accessible, high-quality data, leading to suboptimal accuracy. Simultaneously, centralized training architectures fail to address privacy concerns in healthcare while maintaining high accuracy. To overcome this, our objective was to develop a collaborative Federated Learning (FL) module, as shown in Fig 1, that combines high-accuracy, privacy-preserving algorithms, and optimized architectures for efficient analysis of sensitive medical data.